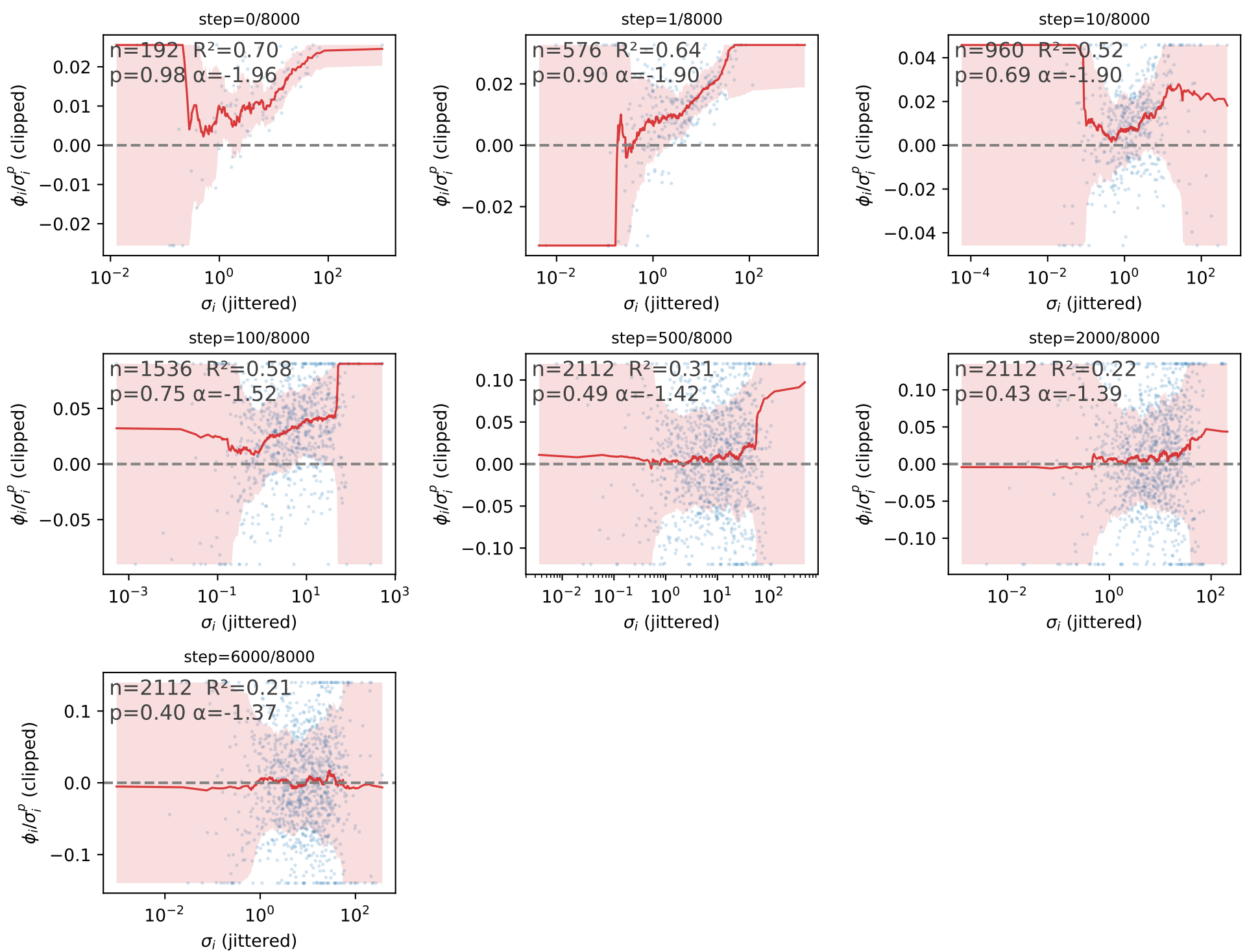


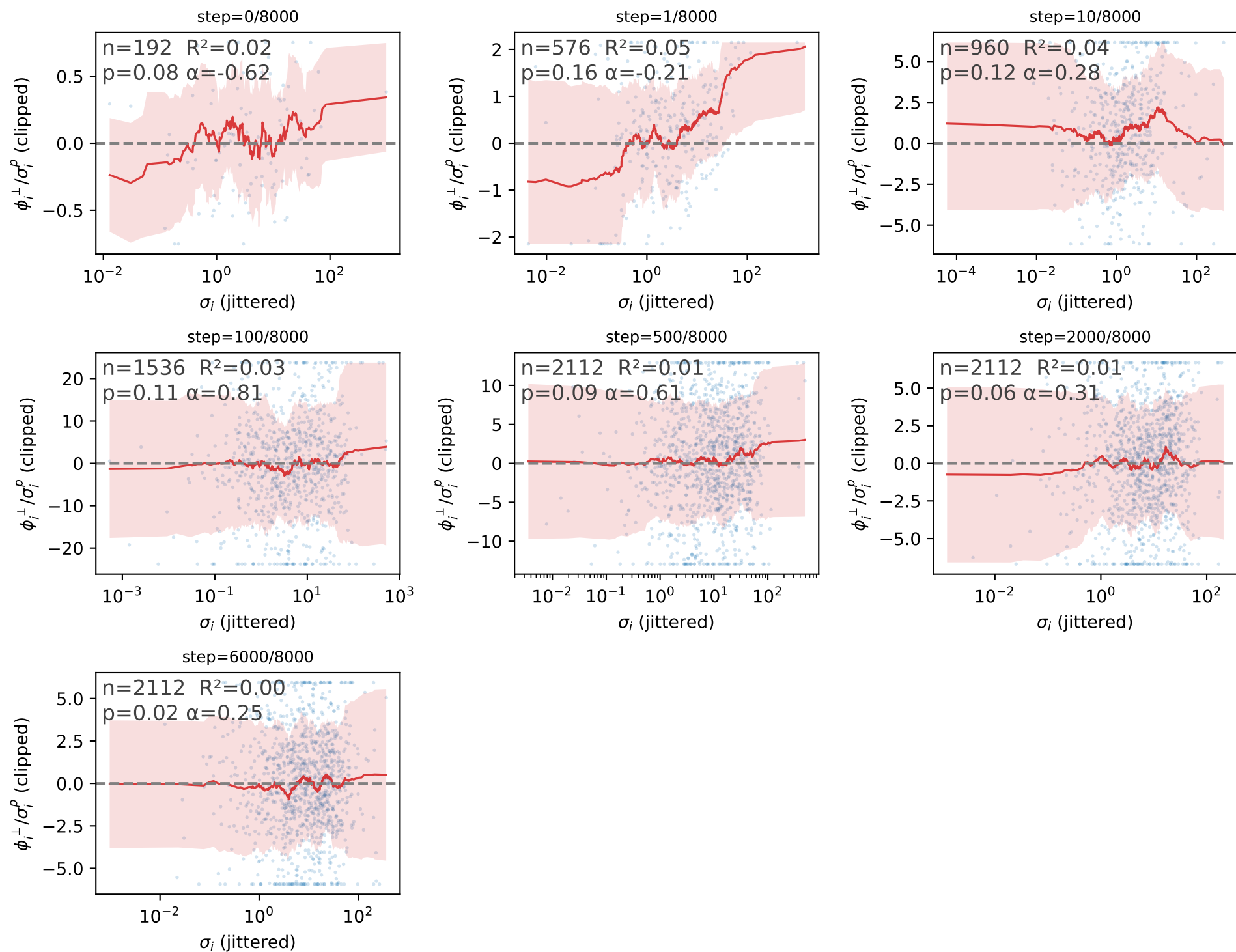
Attention value projection: ϕ_i/σ_i^p vs σ_i (p fit per group from $\log|\phi_i| \sim p\log\sigma_i$, one batch-mean point per mode per step in the group)



n: number of points.

R^2 , p , α : least-squares fit $\log_{10}(y) = p \cdot \log_{10}(x) + \alpha$ (p the fitted power-law exponent, α the intercept).

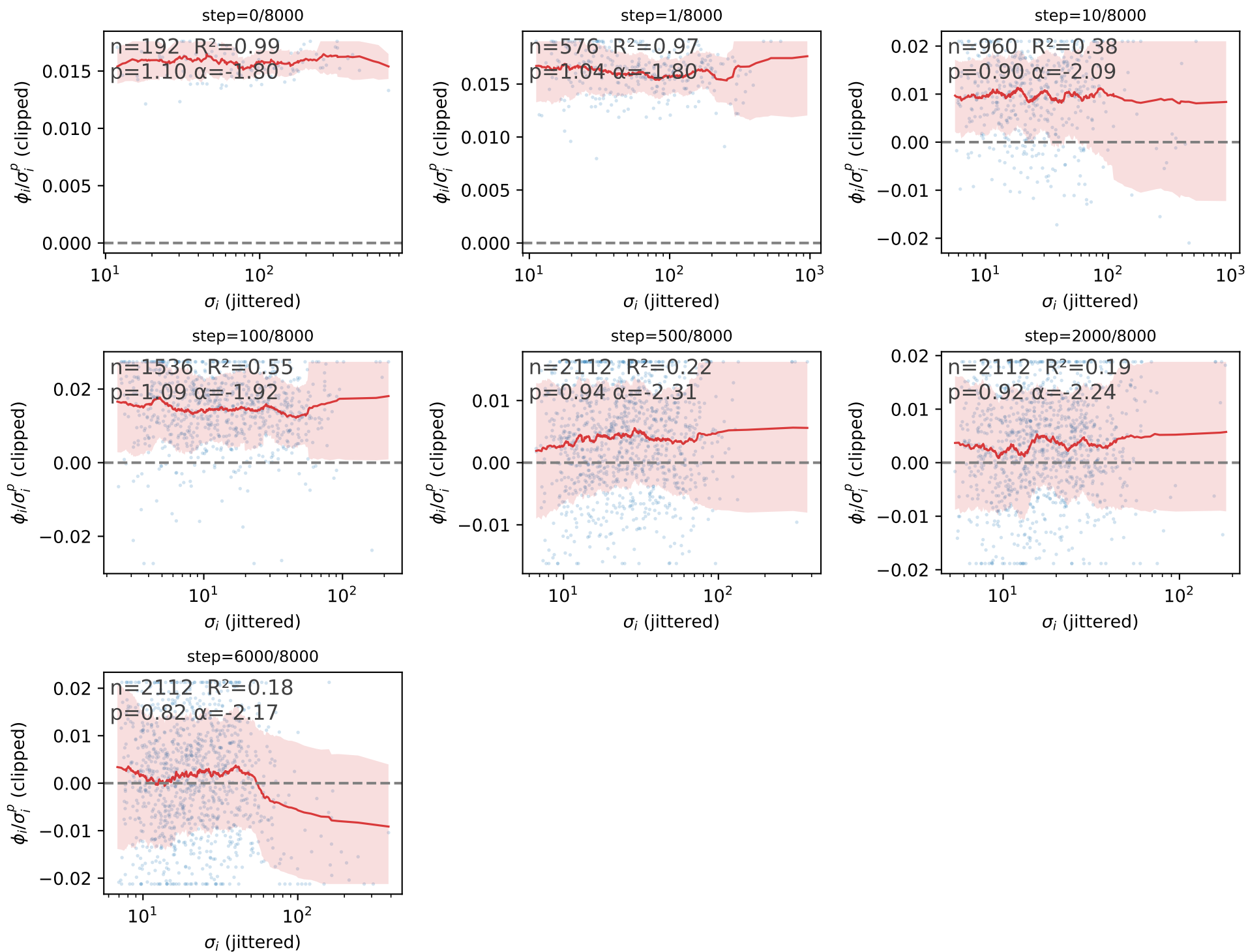
Attention value projection: $\phi_i^\perp/\sigma_i^p$ vs σ_i (p fit per group from $\log|\phi_i^\perp| \sim p \log \sigma_i$, one batch-mean point per mode per step in the group)



n: number of points.

R^2 , p , α : least-squares fit $\log_{10}(y) = p \cdot \log_{10}(x) + \alpha$ (p the fitted power-law exponent, α the intercept).

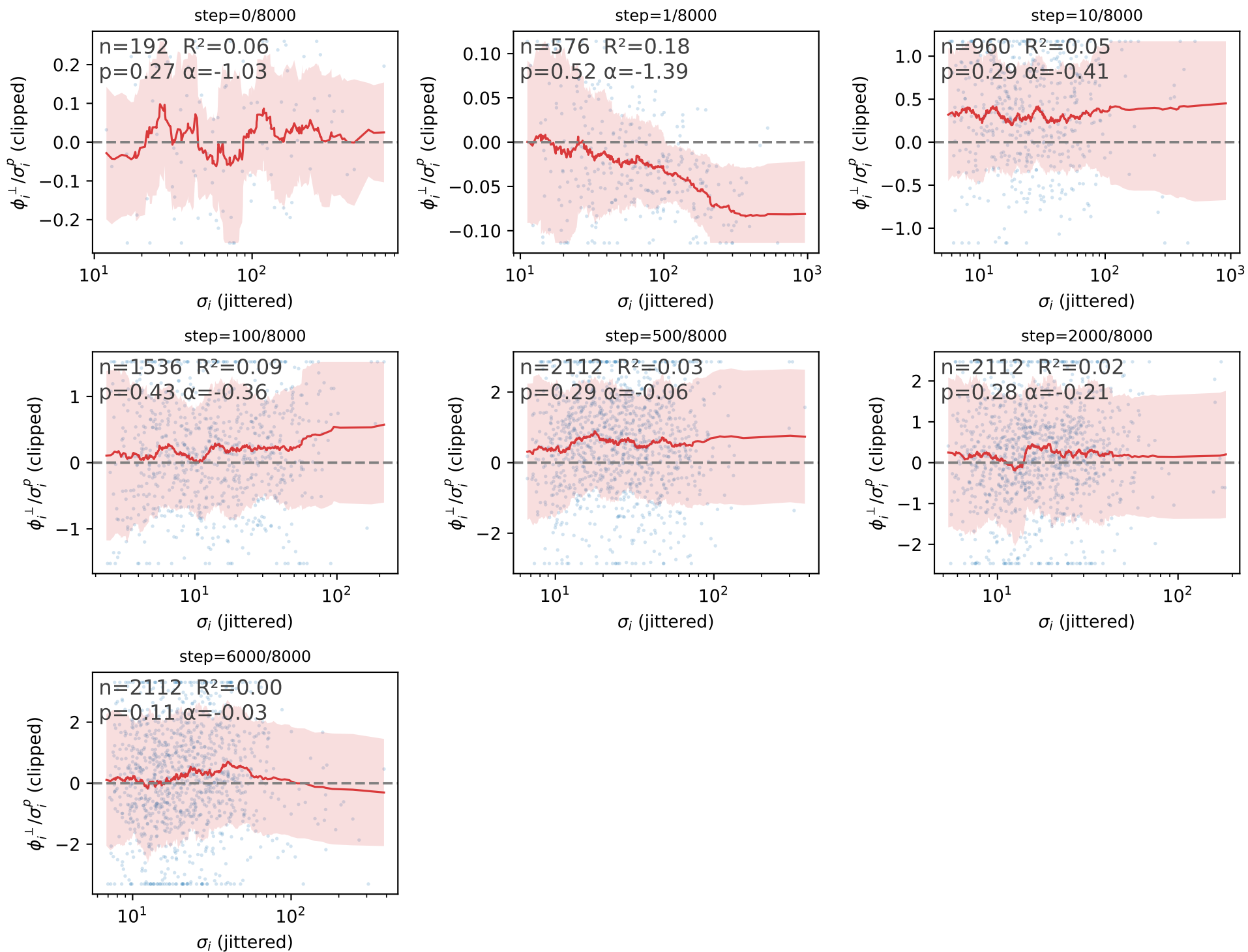
MLP up-projection: ϕ_i/σ_i^p vs σ_i (p fit per group from $\log|\phi_i| \sim p \log \sigma_i$, one batch-mean point per mode per step in the group)



n : number of points.

R^2 , p , α : least-squares fit $\log_{10}(y) = p \cdot \log_{10}(x) + \alpha$ (p the fitted power-law exponent, α the intercept).

MLP up-projection: $\phi_i^\perp/\sigma_i^p$ vs σ_i (p fit per group from $\log|\phi_i^\perp| \sim p \log \sigma_i$, one batch-mean point per mode per step in the group)



n: number of points.

R², p, α : least-squares fit $\log_{10}(y) = p \cdot \log_{10}(x) + \alpha$ (p the fitted power-law exponent, α the intercept).